

Notes for Colloquium

Combining the strengths of Rubin, Euclid, and Roman (and radio surveys!) for time-domain science and cosmology
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LSST at Rubin Observatory (2025-2035)

- optical observation across 6 dif bands

Euclid

- powerful for weak lensing

Romans Telescope

- wide area deep (in infrared)

• The detection and deblending challenges for Rubin galaxies

- so many galaxies that they overlap

- they use other telescopes to overlap Rubin

- They model Rubin & Euclid galaxies together to detect spectrometry

Joint Analysis for transient

Parameter

- Want the STD of host nuclei

- Tidal disruption events, look at multiple supernova

ERDDP

- ground based data

- Making Rubin more useful to see shapes of galaxies better

DDP3: Joint detection & pixel modeling

- have model for pixel scale

- making model to look like each observation

- Use for dark energy

training model

Challenges for Multi-resolution model

- Don't count for complex shaped galaxies

- Used every pixel for a model stabilized by ML

Trained to see

- Scarlet 2:

- written in Jax, fully differentiable



- Promolal operators to stabilize fitting

Wavelet decomposition for multi-resolution
Template-free transient photometry

Testing joint processing of Rubin & Euclid

- Use simulations to see how well they did

- Joint analysis improved Blend

The Rubin - E DRP3 pipeline

- 1. Resample images

 - model and frame

 - Correlated noise

 - power spectrum

Precise SNIa photometry: testing with DES sequence

- Test simulation

- Use Radio size to find tidal transition zone

Why was Scarlet2 written in Jax?

What other ML systems are able to fit multiple parameters on the scale of a parameter for each pixel in a Rubin image?

Does Scarlet2 use Gaussians to fit the model?